Region Growing and Region Splitting

Region Growing and Region Splitting

Edges and thresholds sometimes do not give good results for segmentation.

Region-based segmentation is based on the connectivity of similar pixels in a region.

There are two main approaches to region-based segmentation: region-growing and region-splitting.

Region Growing and Region Splitting

Procedure for Region-Growing

1) Find all connected components in $S(x,y)$ and reduce each connected component to one pixel. Label all such pixels found as 1. All other pixels in S are labeled 0.

2) Form an image f_Q such that, at each point (x,y), $f_Q(x,y) = 1$ if the input image satisfies a given predicate, Q, at those coordinates, and $f_Q(x,y) = 0$ otherwise.

Region Growing and Region Splitting

3) Let g be an image formed by appending to each seed point in S all the 1-valued points in f_Q that are 8-connected to that seed point.

4) Label each connected component in g with a different region label (Ex. integers or letters). This is the segmented image obtained by region growing.

Region Growing and Region Splitting

Procedure for Region splitting and Merging

1) If a region R is inhomogeneous ($P(R) = \text{False}$), then R is split into four sub-regions.

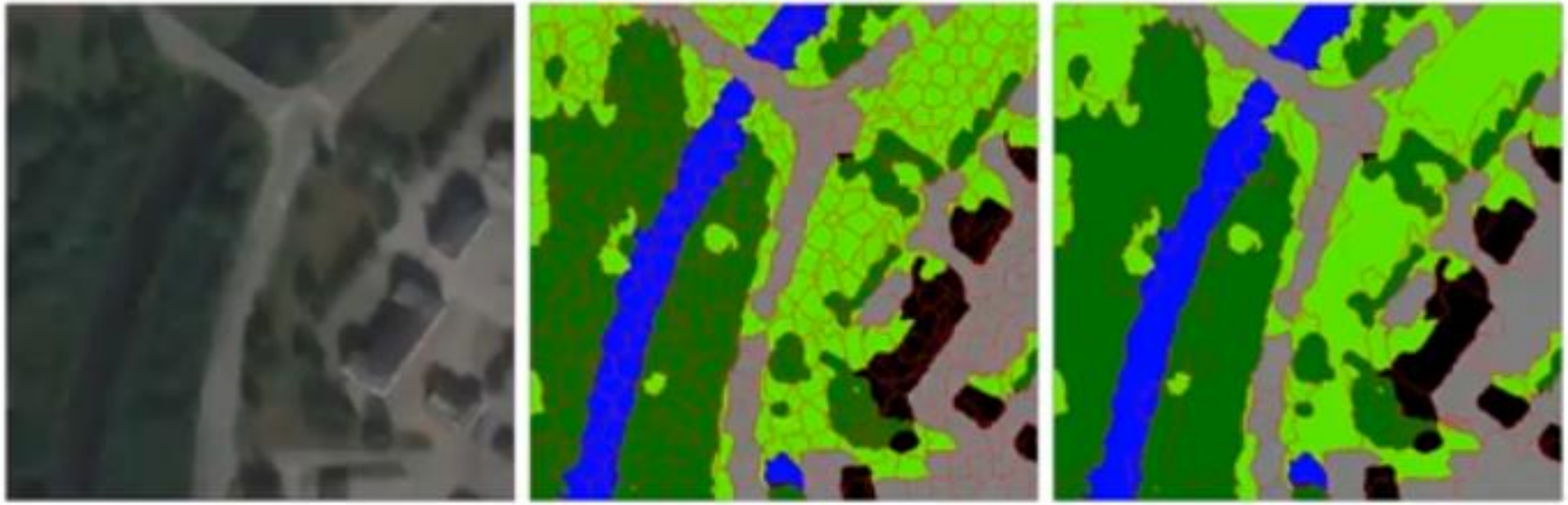
2) If 2 adjacent regions R_i, R_j are homogeneous ($P(R_i \cup R_j) = \text{True}$), then they are merged.

3) The algorithm stops when no further splitting or merging is possible.

Note : Condition for region growing :
abs(seed value − pixel value) $\leq$ Threshold

Region Merging/Growing

Used to segment an image into regions of similar intensity or color. The algorithm is used to evaluate the values within a regional span and grouped together based on the **merging** criteria



Region Growing

Q1. Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.

$T = 2$

$$\begin{array}{c|cccc|} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 1 & 2 & 0 \\ 1 & 2 & 5 & 6 & 1 \\ 2 & 1 & 4 & 7 & 3 \\ 3 & 0 & 2 & 5 & 1 \\ \hline \end{array}$$

Ans. The segmented region is shown in the following figure.

Condition $\rightarrow$ absolute difference $<= 2$.

4 way connectivity

Region Growing

Q1. Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.

	0	1	2	3
0	0	1	2	0
1	2	5	6	1
2	1	4	7	3
3	0	2	5	1

$T = 2$

a

$7 - 6 = 1$ ✓

$7 - 4 = 3$ ✗

$7 - 5 = 2$ ✓

$7 - 3 = 4$ ✗

Ans. The segmented region is shown in the following figure.

Condition → absolute difference $<= 2$

4 way connectivity

Region Growing

Q1. Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.

T = 2

	0	1	2	3
0	0	1	2	0
1	2	5^a	6^a	1
2	1	4	7	3
3	0	2	5^a	1

a

7 − 6 = 1 ✓
7 − 4 = 3 ✗
7 − 5 = 2 ✓
7 − 3 = 4 ✗

Ans. The segmented region is shown in the following figure.

Condition → absolute difference <= 2.

4 way connectivity

Region Growing

Ans. The segmented region is shown in the following figure.

Condition → absolute difference <= 2.

4 way connectivity

Here, we will compare each element which is 4 way connected to the seed element.

	0	1	2	3
0	0	1	2	0
1	2	5^a	6^a	1
2	1	4	7^a	3
3	0	2	5^a	1

Region Growing

Segmented image obtained from region growing:

	0	1	2	3
0	0	0	0	0
1	0	1	1	0
2	0	0	1	0
3	0	0	1	0

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

Seed point = 6
T = 3

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Condition → absolute difference <= 3.
8 way connectivity.

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Seed point = 6

T = 3

a

Condition → absolute difference <= 3.

8 way connectivity.

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

5^a	6^a	6^a	7^a	6	7^a	6^a	6^a
6^a	7^a	6^a	7^a	5^a	5^a	4^a	7^a
6^a	6^a	4^a	4^a	3	2	5^a	6^a
5^a	4^a	5^a	4	2	3	4^a	6^a
0	3	2	3	3	2	4^a	7^a
0	0	0	0	2	2	5^a	6^a
1	1	0	1	0	3	4^a	4^a
1	0	1	0	2	3	5	4

Seed point = 6

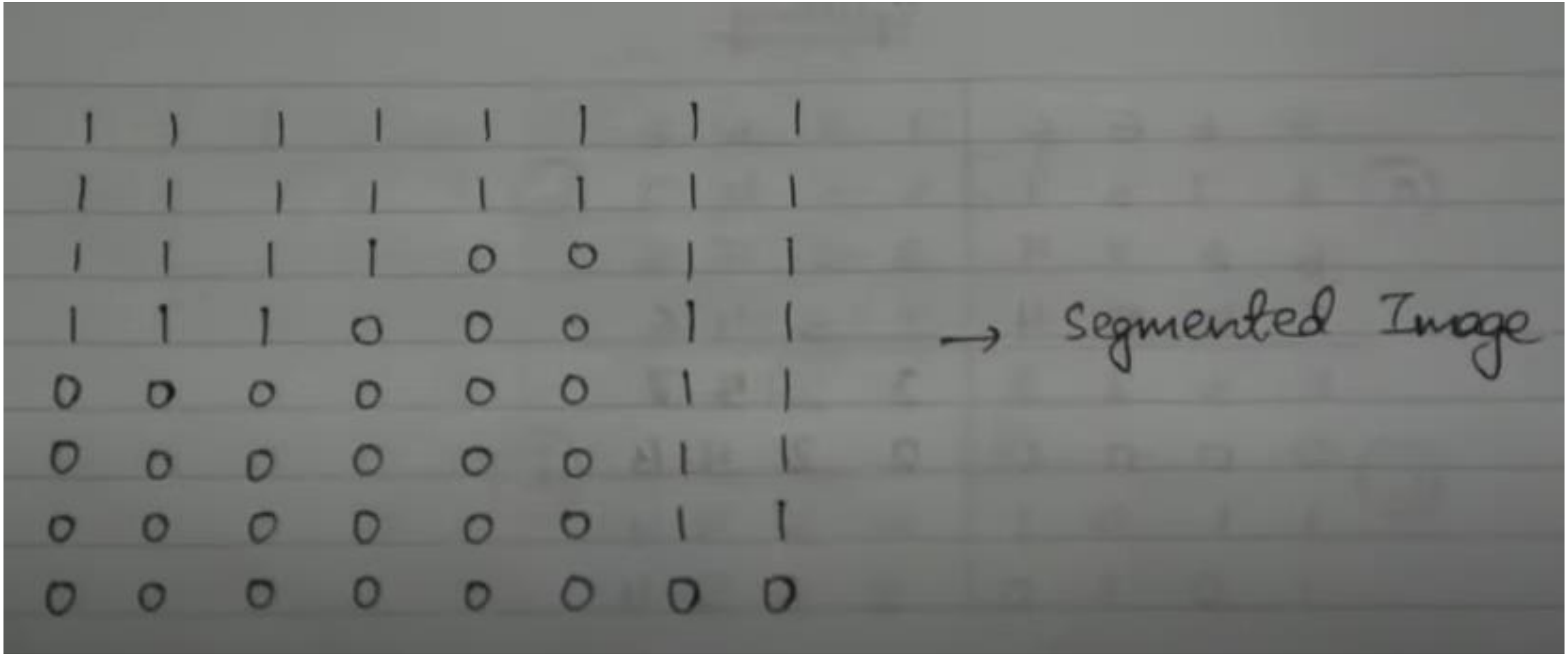
T = 3

a

Condition → absolute difference <= 3.

8 way connectivity.

Region Growing



Region Growing

Region Growing Segmentation

Example: Consider a 3-bit image given below. Assuming seed value of 6, segment the following image using 8-connectivity and threshold=3

How many segments are identified ?

Ans: **Conditions for growing a region**

8-Connectivity between seed pixel and given pixel

abs(seed_value - pixel_value) < 3

seed= 6

2	2	7	2	1
1	7	6	6	2
7	6	6	5	7
2	4	5	4	2
1	2	5	1	1

Aplying thresholding condition

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Region Growing

Aplying thresholding condition

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Aplying 8-connectivity to label each pixel

a	a	b	c	c
a	b	b	b	c
b	b	b	b	b
d	b	b	b	e
d	d	b	e	e

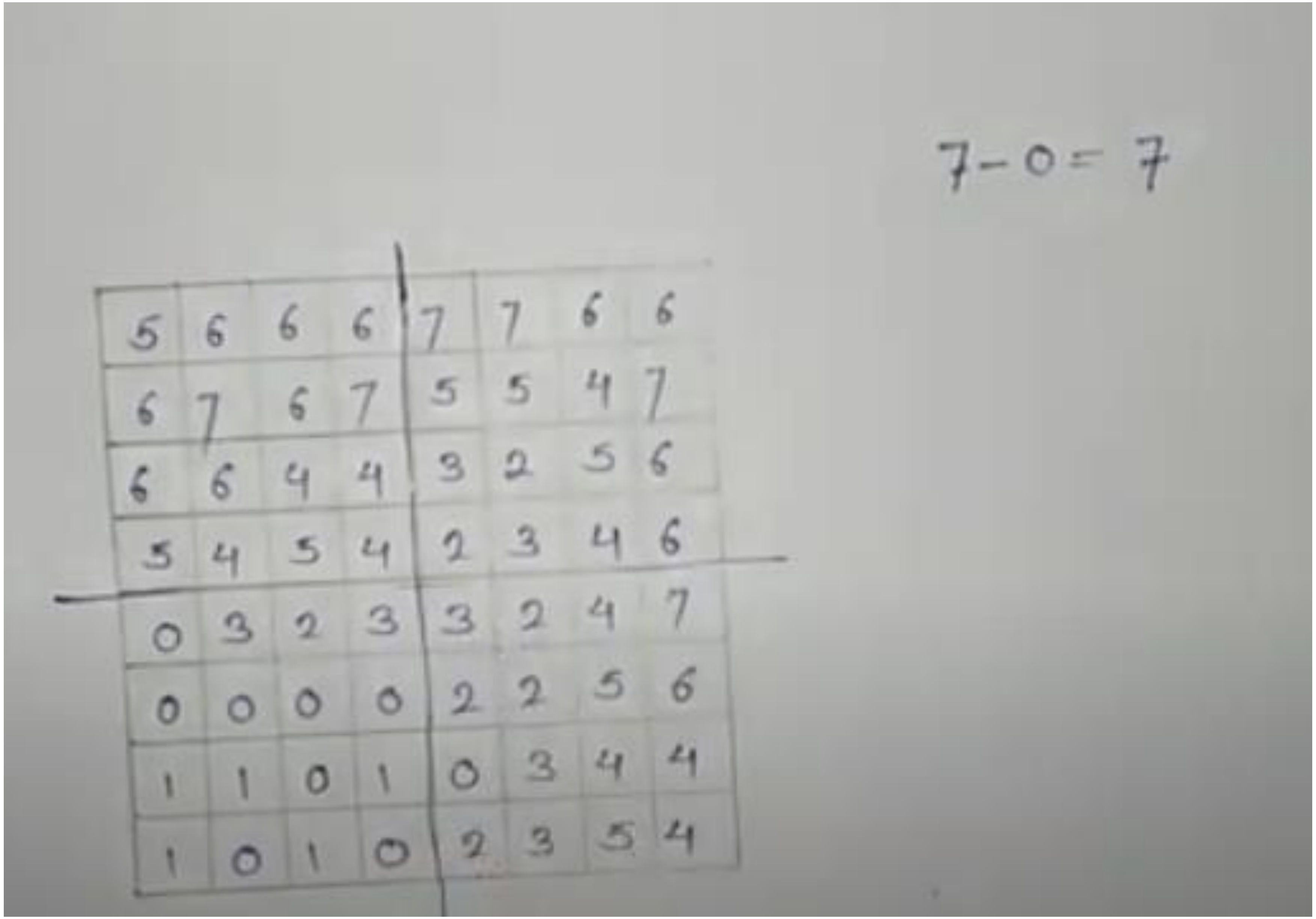
Region Splitting

Q. Apply the region splitting on the following image

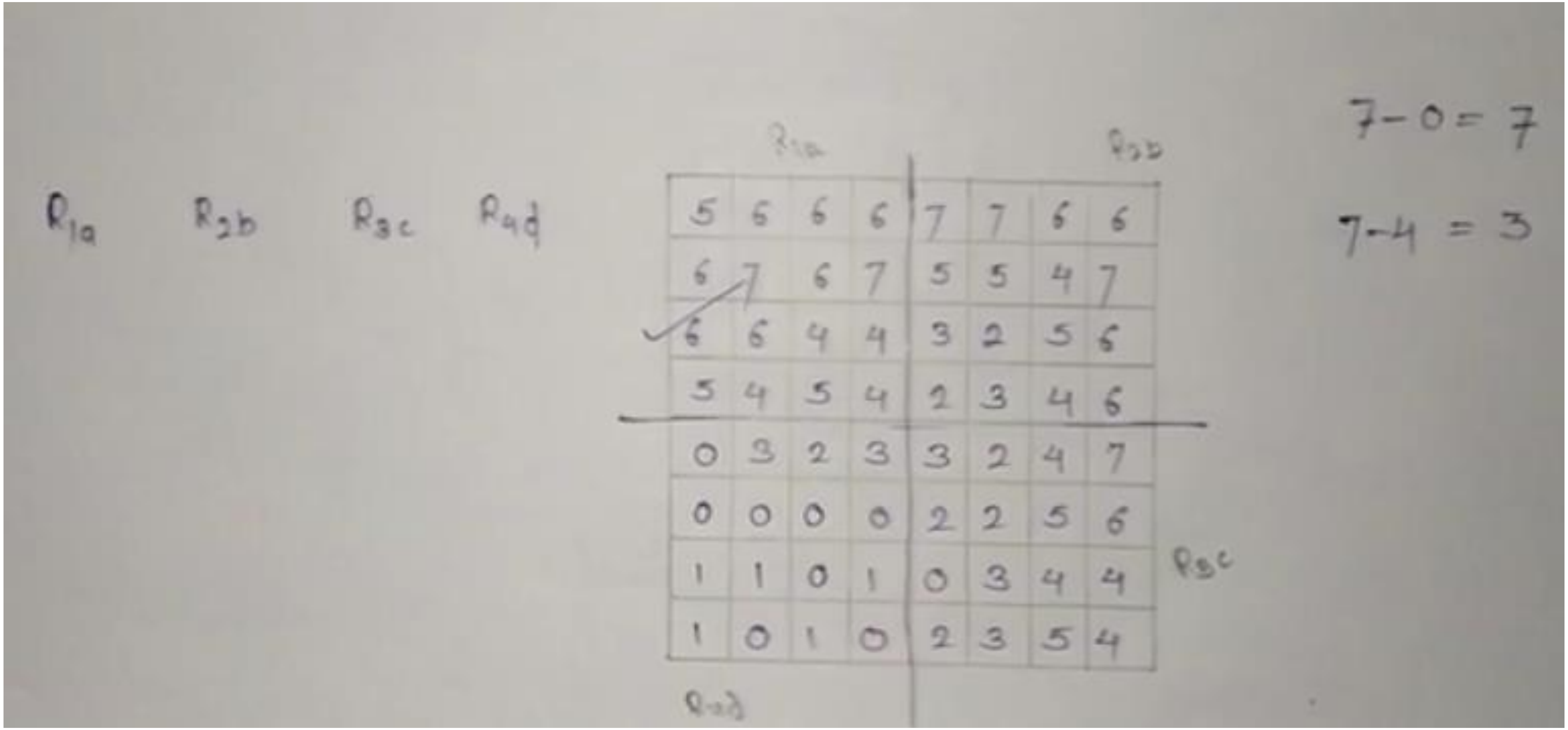
Assume the threshold value be ≤ 4

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

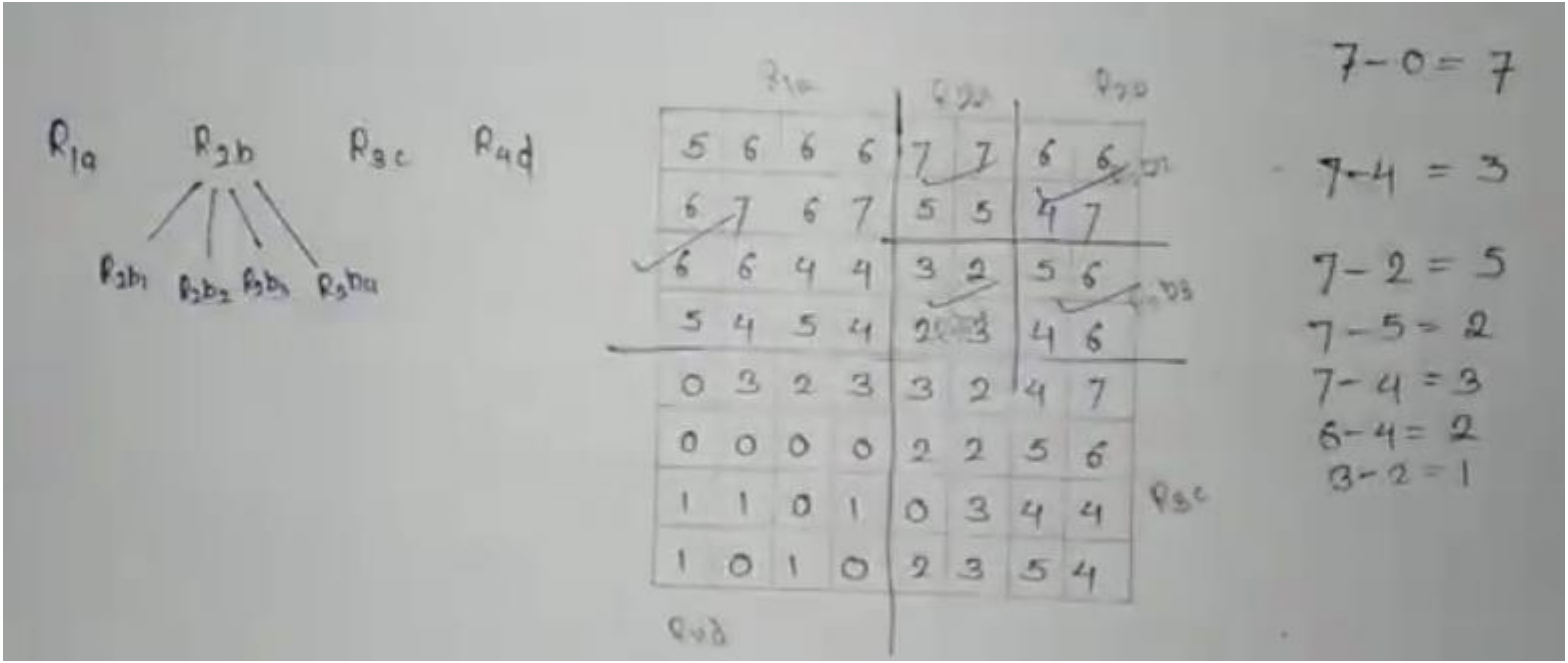
Region Splitting



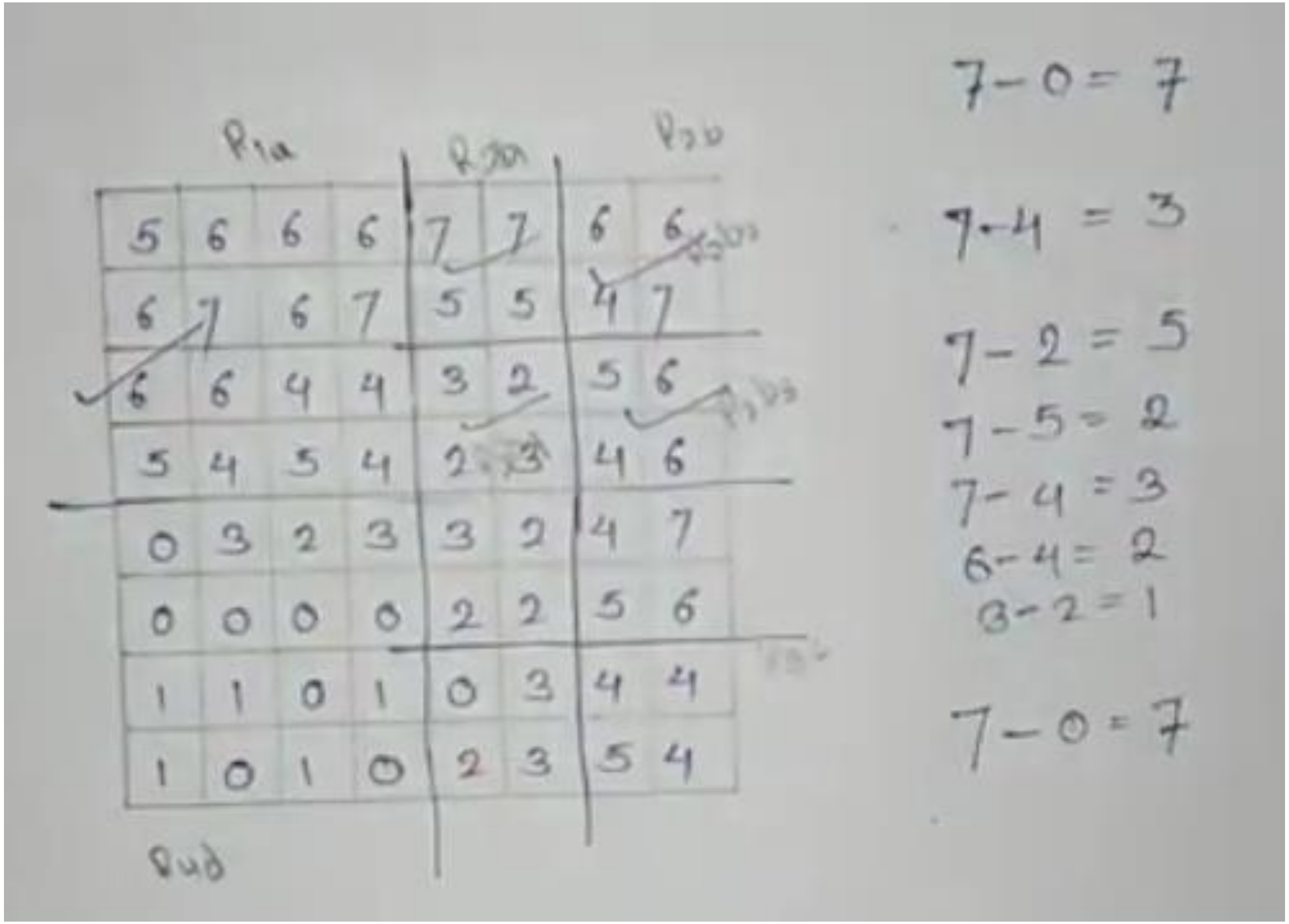
Region Splitting



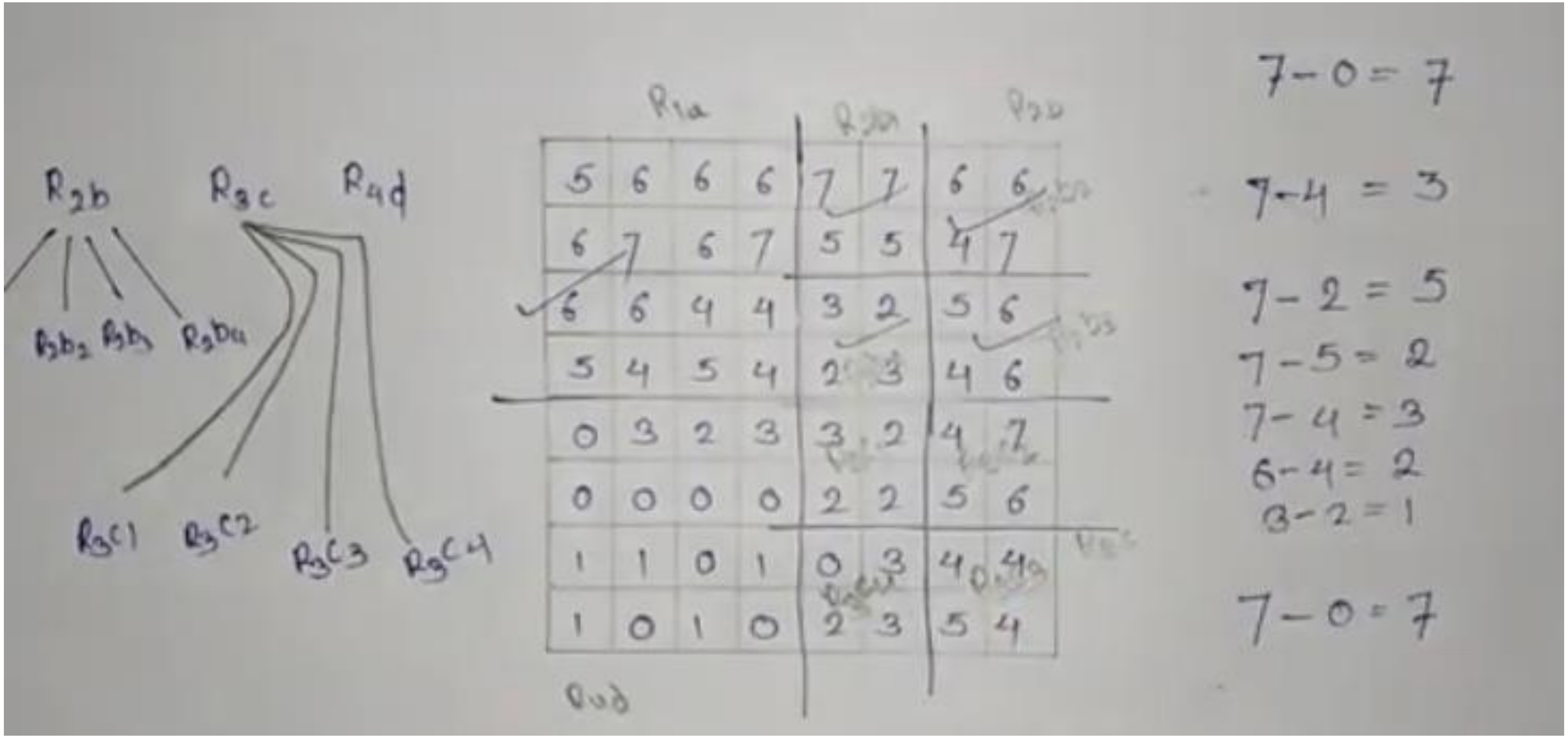
Region Splitting



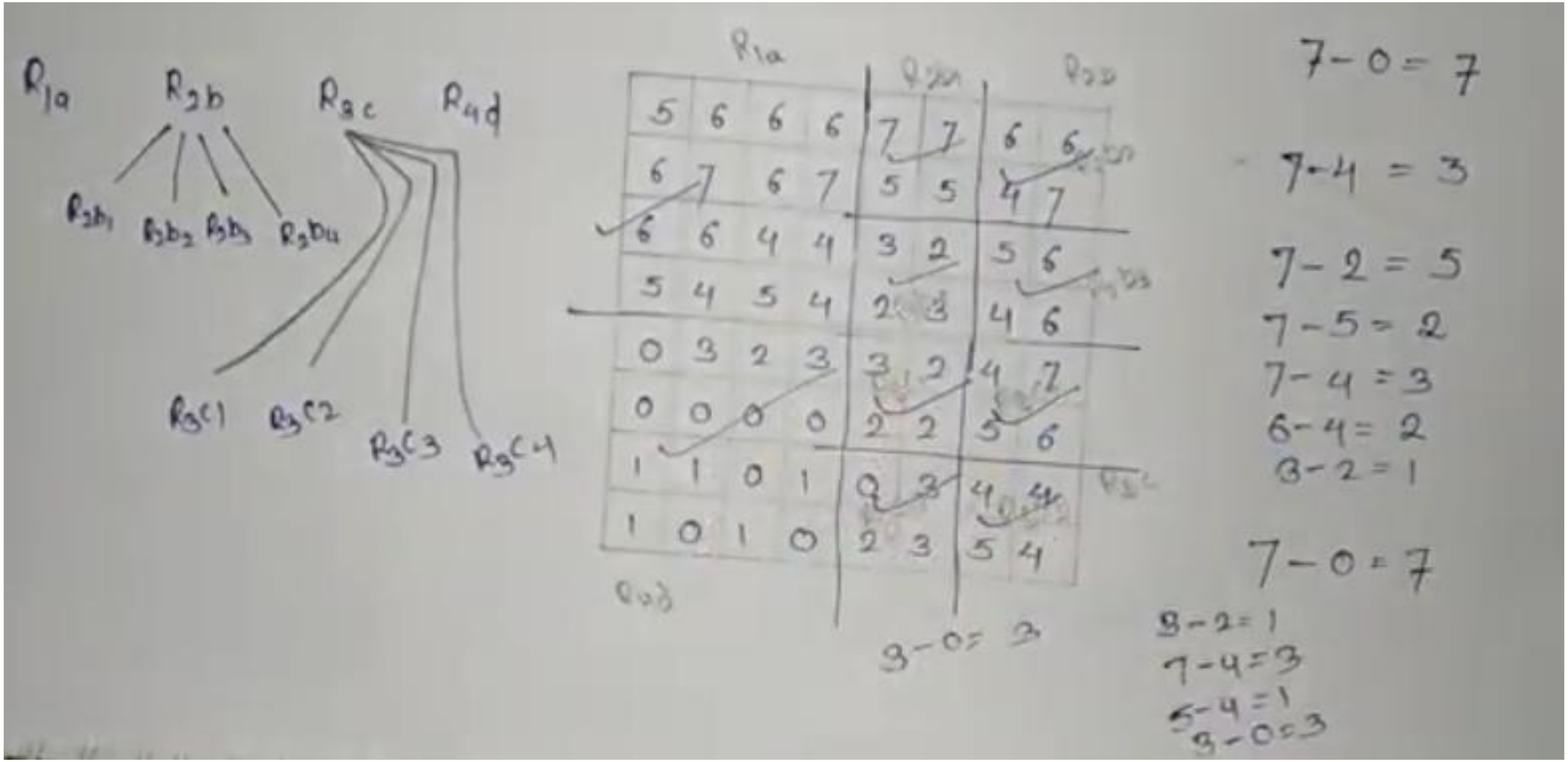
Region Splitting



Region Splitting



Region Splitting



Region splitting and Merging

Q3. Apply splitting and merging on the following image with threshold value equal to 3.

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Ans. Condition : absolute difference <= 3.

Max value = 7

Min value = 0

$|7-0| = 7$ which is greater than 3.

Therefore we will split the region into 4 sub-regions.

Region splitting and Merging

Splitting

	5	6	6	6	7	7	6	6	
a	6	7	6	7	5	5	4	7	b
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	5	7	
c	0	0	0	0	2	2	5	6	d
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

In region a :

Max = 7 Min = 4

$|7-4| = 3$ which is equal to threshold.

Therefore, no need to split.

Region splitting and Merging

Splitting

	5	6	6	6	7	7	6	6	
(a)	6	7	6	7	5	5	4	7	(b)
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	5	7	
(c)	0	0	0	0	2	2	5	6	(d)
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

$7 - 2 = 5$

Region splitting and Merging

Splitting

a	5	6	6	6	7	7	6	6	b
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
c	0	3	2	3	3	2	5	7	d
	0	0	0	0	2	2	5	6	
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

$|7-2| = 5$

$|3-0| = 3$

$|7-0| = 7$

Region splitting and Merging

In region (b) :

Max = 7 Min = 2

$|7-2| = 5$ which is greater than 3.

Therefore we will split the region (b) into 4 sub-regions.

In region (c) :

Max = 3 Min = 0

$|3-0| = 3$.

So no need to split.

In region (d) :

Max = 7 Min = 0

$|7-0| = 7$.

So we will split into 4 sub-regions.

Region splitting and Merging

					b1		b2		
	5	6	6	6	7	7	6	6	
a	6	7	6	7	5	5	4	7	b
	6	6	4	4	3	2	5	6	
					b3		b4		
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	5	7	
					d1		d2		
c	0	0	0	0	2	2	5	6	d
	1	1	0	1	0	3	4	4	
					d3		d4		
	1	0	1	0	2	3	5	4	

Furthermore, we will check all the sub-regions. Since all of them are <= 3, no further splitting is required.

Region splitting and Merging

Merging

Check adjacent regions, if they are falling within the threshold, then merge.

Consider regions (a) and (b1)

Max = 7 Min = 4

|7-4| = 3. ✓ Merge.

Region splitting and Merging

					b1		b2		
	5	6	6	6	7	7	6	6	
a	6	7	6	7	5	5	4	7	b
	6	6	4	4	3 b3	2	5 b4	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3 d1	2	5 d2	7	
c	0	0	0	0	2	2	5	6	d
	1	1	0	1	0 d3	3	4 d4	4	
	1	0	1	0	2	3	5	4	

Furthermore, we will check all the sub-regions. Since all of them are <= 3, no further splitting is required.

Merging

Check adjacent regions, if they are falling within the threshold, then merge.

Consider regions a and b1
Max = 7 Min = 4
|7-4| = 3 ✓ Merge.

Region splitting and Merging

Consider regions a b1 and b2

Max = 7 Min = 4

$|7-4| = 3$. ✓ Merge.

Consider regions a b1 b2 and b4

Max = 7 Min = 4

$|7-4| = 3$ ✓ Merge

Consider regions a b1 b2 b4 and d2

Max = 7 Min = 4

$|7-4| = 3$ ✓ Merge.

Similarly, merge a b1 b2 b4 d2 with d4.

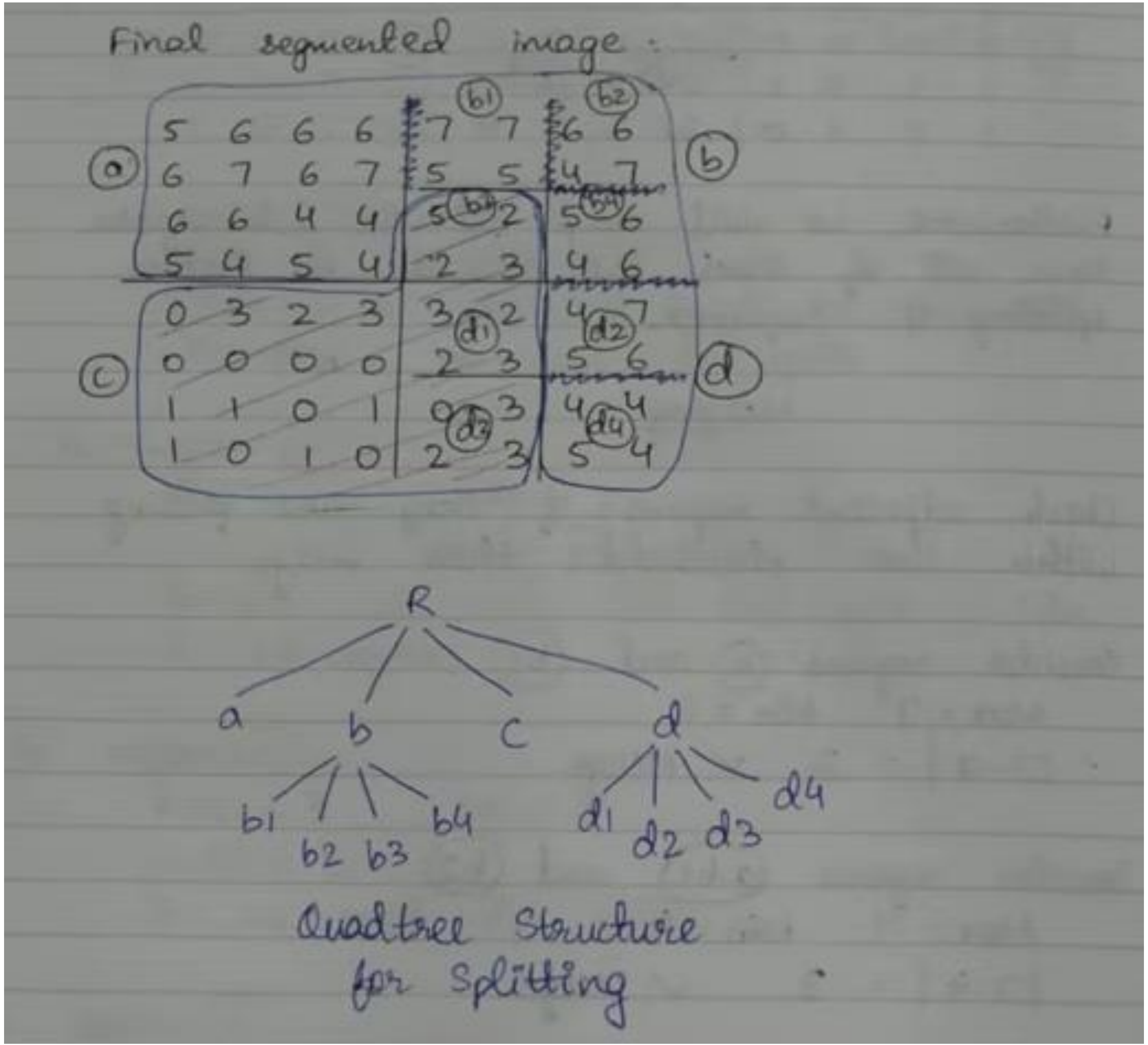
Region splitting and Merging

Similarly, merge (c), (d1), (b3) and (d3).

Final segmented image :

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	3	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region splitting and Merging



Thank you